Mathematical Modeling

Chapter 3: The Simplex Method

Personal study notes

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1 The problem and its possible outcomes

The primal simplex method solves a linear program by maintaining a feasible basis and exchanging one basic variable at a time. For a maximization problem, use the standard form

$$\max c^T x, \qquad Ax = b, \qquad x \geq 0, \quad A \in \mathbb{R}^{m \times n},\ b \in \mathbb{R}^m,\ c, x \in \mathbb{R}^n. \tag{1}$$

Unless explicitly discussing redundant rows, assume $\operatorname{rank} A = m$ and $m \leq n$. If $m = n$, the nonsingular system has a single candidate $x = A^{-1}b$: check its nonnegativity directly. The interesting basis exchanges occur when $m < n$. Minimization can be handled by maximizing the negative objective and reversing the sign of the final value.

There are three main outcomes. An infeasible problem has no point satisfying all constraints. A feasible problem with objective unbounded above has no finite maximum. Otherwise the maximum is finite and attained; in standard form at least one optimal solution is a vertex of the feasible set. The optimal solution may be unique or there may be infinitely many: the segment between two distinct optimal points is also optimal.

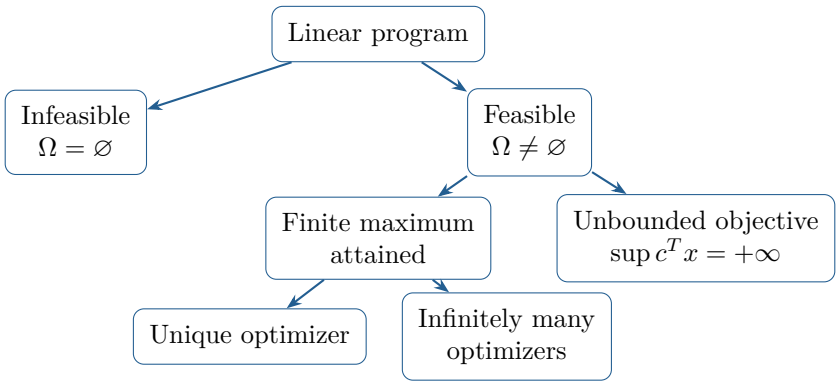


Figure 1: Classify feasibility first, then objective behavior. An unbounded feasible set can still have a finite maximum.

For example, $\max(-x)$ over $x \geq 0$ has an unbounded feasible set but the unique optimizer $x = 0$. Unboundedness of the set and of the objective are separate questions.

2 Bases, basic solutions, and degeneracy

Write $A = [a_1 \ \cdots \ a_n]$, where $a_j \in \mathbb{R}^m$ is column j of the original constraint matrix A. Let $B = (j_1, \ldots, j_m)$ be an ordered list of indices whose columns form the nonsingular matrix $A_B = [a_{j_1} \ \cdots \ a_{j_m}]$. List the remaining indices as $N = (k_1, \ldots, k_{n-m})$ and define

$$A_N = [a_{k_1} \ \cdots \ a_{k_{n-m}}], \qquad x_B = \begin{pmatrix} x_{j_1} \\ \vdots \\ x_{j_m} \end{pmatrix}, \qquad x_N = \begin{pmatrix} x_{k_1} \\ \vdots \\ x_{k_{n-m}} \end{pmatrix}.$$

Order c_B and c_N by the same indices. Then

$$A_B x_B + A_N x_N = b, \qquad x_B = A_B^{-1} b - A_B^{-1} A_N x_N.$$

Define the basic solution associated with B as the vector x obtained by setting $x_N = 0$ and solving $A_B x_B = b$, so $x_B = A_B^{-1} b$. It is a *basic feasible solution* (BFS, Polish BRD) exactly when $x_B \geq 0$. A feasible basic solution is *degenerate* when at least one basic coordinate is zero. A nonbasic zero does not establish degeneracy.

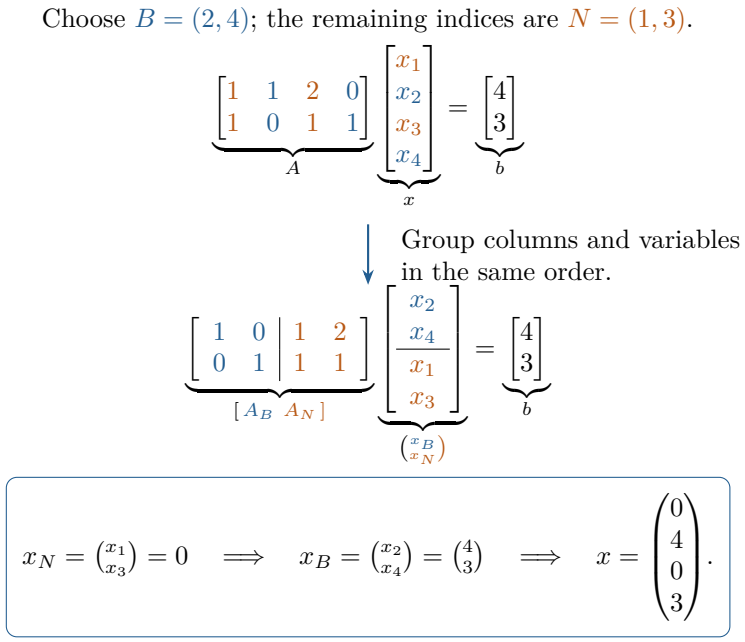


Figure 2: An illustrative partition with $m = 2$, $n = 4$. Blue marks the basic columns and variables; orange marks the nonbasic ones. Here $A_B = I_2$, so setting $x_N = 0$ immediately gives a nondegenerate BFS. Grouping changes the display order, not the original variable numbering.

A BFS is a vertex of $\{x : Ax = b, x \geq 0\}$, and conversely every vertex admits a feasible basis. Different bases can represent the same degenerate vertex. There are at most $\binom{n}{m}$ bases, but some column subsets are singular, some basic solutions are infeasible, and some solutions coincide. This is an upper bound, not an exact count of vertices.

2.1 Classifying three candidate solutions

Task statement. For the following system, determine whether each supplied vector is a basic solution and identify its basic variables. Then check feasibility under $x \geq 0$ and, for a BFS, degeneracy:

$$\begin{aligned} 2x_1 + x_2 + 3x_3 + x_6 &= 8, \\ x_1 + 2x_3 + x_5 + x_6 &= 5, \\ 3x_1 + x_3 + x_4 + 2x_6 &= 4. \end{aligned} \tag{2}$$

$$(0, 8, 0, 4, 5, 0)^T, \qquad (0, \tfrac{1}{2}, \tfrac{5}{2}, \tfrac{3}{2}, 0, 0)^T, \qquad (0, 0, 0, -12, -3, 8)^T.$$

The columns of A and the right-hand side are:

Equation	a_1	a_2	a_3	a_4	a_5	a_6	b
1	2	1	3	0	0	1	8
2	1	0	2	0	1	1	5
3	3	0	1	1	0	2	4

Here A has full row rank because columns $(2, 5, 4)$ form I_3. Solving the indicated three-column systems gives:

Candidate x^T	Ordered basis B	Basic values x_B^T	Classification
$(0, 8, 0, 4, 5, 0)$	$(2, 5, 4)$	$(8, 5, 4)$	Nondegenerate BFS
$(0, \frac{1}{2}, \frac{5}{2}, \frac{3}{2}, 0, 0)$	$(2, 3, 4)$	$(\frac{1}{2}, \frac{5}{2}, \frac{3}{2})$	Nondegenerate BFS
$(0, 0, 0, -12, -3, 8)$	$(4, 5, 6)$	$(-12, -3, 8)$	Basic, infeasible

For the second candidate, $2x_3 = 5$, then $x_2 + 3x_3 = 8$ and $x_3 + x_4 = 4$. For the third, $x_6 = 8$, $x_5 = 5 - 8 = -3$, and $x_4 = 4 - 16 = -12$. Each listed basis is nonsingular. Checking $Ax = b$ alone cannot establish feasibility; conversely, counting nonzero entries alone cannot establish that an arbitrary solution is basic without checking column independence.

3 Canonical equations and reduced costs

For a chosen feasible basis, define

$$\bar{A} = A_B^{-1} A, \qquad \bar{b} = A_B^{-1} b \geq 0. \tag{3}$$

The basic columns of $\bar{A}$ form the identity. The resulting canonical equations are

$$x_B + \bar{A}_N x_N = \bar{b}.$$

This is canonical *relative to the basis*: any nonsingular basis produces identity columns after transformation, but only a feasible basis gives $\bar{b} \geq 0$. In numerical code solve systems with A_B rather than computing its inverse explicitly.

Substitute the canonical equations into the objective:

$$\begin{aligned} c^T x &= c_B^T x_B + c_N^T x_N \\ &= c_B^T \bar{b} + (c_N^T - c_B^T \bar{A}_N) x_N \\ &= z_0 - \sum_{j \in N} q_j x_j, \qquad z_0 = c_B^T \bar{b}, \quad q_j = c_B^T \bar{a}_j - c_j. \end{aligned} \tag{4}$$

The indicators $q_j = z_j - c_j$ use the *maximization* sign convention. Another common convention calls $\rho_j = c_j - z_j = -q_j$ the reduced cost. For every basic column, $q_j = 0$.

Indicator	Equivalent reduced cost	Effect of increasing x_j
$q_j < 0$	$\rho_j > 0$	Objective increases if a positive step is feasible
$q_j = 0$	$\rho_j = 0$	Objective stays constant along this direction
$q_j > 0$	$\rho_j < 0$	Objective decreases

Proposition 3.1 (Optimality certificate)**.** *If the basis is feasible and $q_j \geq 0$ for every nonbasic index, then its basic solution is globally optimal.*

Proof. Every feasible point obeys the same transformed equations. Since its nonbasic coordinates are nonnegative, Equation (4) gives $c^T x \leq z_0$. The current BFS attains z_0. □

For a nondegenerate BFS the condition is also necessary: if $q_k < 0$, all basic coordinates are positive, so a sufficiently small positive step in x_k preserves nonnegativity and strictly improves the objective. At a degenerate BFS that step can be blocked at zero. For example, $\max x_2$ subject to $x_1 + x_2 = 0$, $x \geq 0$ has just one feasible point, $(0, 0)$. For $B = (1)$, $q_2 = -1$, yet the point is optimal. Changing to $B = (2)$ gives $q_1 = 1$ and a certificate. Optimality of a point and the sign certificate of a particular basis must be distinguished.

If all nonbasic $q_j > 0$, equality in the certificate forces $x_N = 0$ and hence $x_B = \bar{b}$: the optimizer is unique. This is sufficient, not necessary. If some nonbasic $q_k = 0$, a *positive feasible step* in that direction produces distinct optimal points. A zero indicator by itself is insufficient: $\max 0$ with $x_1 + x_2 = 0$, $x \geq 0$ has $q_2 = 0$ and still only one feasible point.

4 One simplex step

4.1 Entering variable and feasible direction

Choose a nonbasic k with $q_k < 0$. Dantzig's rule chooses a most negative indicator; any negative indicator is eligible. Hold other nonbasic coordinates at zero and increase $x_k = \theta \geq 0$. With $u = \bar{a}_k$,

$$x_B(\theta) = \bar{b} - \theta u, \qquad d_B = -u, \quad d_k = 1, \quad d_j = 0 \ (j \in N \setminus \{k\}). \tag{5}$$

Then $Ad = 0$ and $c^T d = -q_k > 0$. The equations remain satisfied along $x(\theta) = x^{(B)} + \theta d$; nonnegativity determines the allowed step.

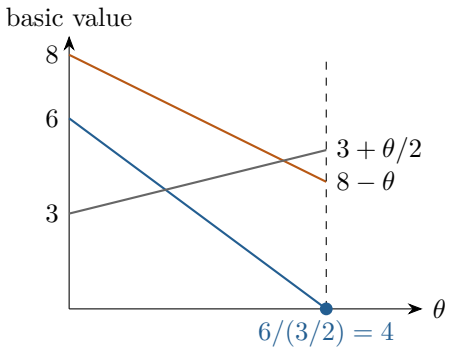
4.2 Ratio test and leaving variable

If $u_i > 0$, basic coordinate x_{j_i} decreases and requires $\theta \leq \bar{b}_i / u_i$. If $u_i \leq 0$, it places no upper bound on the step. Thus

$$\theta^* = \min_{i : u_i > 0} \frac{\bar{b}_i}{u_i}. \tag{6}$$

Choose a minimizing row r and remove its basic variable x_{j_r}. The pivot element is $p = \bar{a}_{rk} > 0$. Replace j_r by k in the ordered basis; $x_k = \theta^*$ and the leaving variable becomes nonbasic and zero.

The objective gain is $-q_k \theta^*$. A tie may leave another zero basic coordinate, making the new BFS degenerate.



Only decreasing coordinates
limit the step.
First zero determines the exit.

Figure 3: Illustrative ratio test: the bounds are 4 and 8; the increasing coordinate gives no bound. The maximum feasible step is $\theta^* = 4$.

4.3 When there is no leaving variable

If $u \le 0$, the minimum in Equation (6) has no candidates. Then $d \ge 0$, so $x^{(B)} + \theta d$ is feasible for all $\theta \ge 0$, and $c^T x(\theta) = z_0 - \theta q_k \to +\infty$. This is a certificate of an unbounded objective. It requires *both* a negative indicator and a column with no positive entry. The same column with $q_k = 0$ instead gives constant objective along a feasible ray.

4.4 Tableau update by row operations

Write a tableau as the constraint rows $[\bar{A} \mid \bar{b}]$, followed by $[q^T \mid z_0]$. The last row expresses $z + \sum_j q_j x_j = z_0$. For a pivot in row r, column k, use old coefficients on the right-hand sides of all update formulas:

$$\begin{aligned} \bar{a}'_{rj} &= \bar{a}_{rj}/p, & \bar{b}'_r &= \bar{b}_r/p, \\ \bar{a}'_{ij} &= \bar{a}_{ij} - \bar{a}_{ik}\bar{a}_{rj}/p, & \bar{b}'_i &= \bar{b}_i - \bar{a}_{ik}\bar{b}_r/p \quad (i \ne r), \\ q'_j &= q_j - q_k\bar{a}_{rj}/p, & z'_0 &= z_0 - q_k\bar{b}_r/p. \end{aligned}$$

Equivalently: divide the pivot row by p, then subtract suitable multiples of that normalized row from the other constraint rows and the indicator row. Column k becomes a unit column, while the unchanged basic columns stay unit columns. The nonzero pivot also guarantees that the replacement basis is nonsingular. Row numbers are *not* variable indices: row r is labelled x_{j_r} before the pivot and x_k afterwards.

5 The algorithm and degenerate steps

1. Start from a feasible basis and form $\bar{A}$, $\bar{b}$, q, and z_0.
2. If all nonbasic indicators are nonnegative, return the current optimum.
3. Select a nonbasic column k with $q_k < 0$.
4. If $\bar{a}_k \le 0$, return an improving feasible ray and unboundedness.
5. Apply the ratio test to positive entries of $\bar{a}_k$; choose a leaving row.
6. Pivot, update the basis and objective row, and return to step 2.

Before: pivot $p = \bar{a}_{rk} > 0$; x_k enters, x_{j_r} leaves.

Basic	$\cdots$	x_j	$\cdots$	x_k	$\cdots$	RHS
$\vdots$		$\vdots$		$\vdots$		$\vdots$
x_{j_i}	$\cdots$	$\bar{a}_{ij}$	$\cdots$	$\bar{a}_{ik}$	$\cdots$	$\bar{b}_i$
$\vdots$		$\vdots$		$\vdots$		$\vdots$
x_{j_r}	$\cdots$	$\bar{a}_{rj}$	$\cdots$	$\boxed{p = \bar{a}_{rk}}$	$\cdots$	$\bar{b}_r$
$\vdots$		$\vdots$		$\vdots$		$\vdots$
q	$\cdots$	q_j	$\cdots$	q_k	$\cdots$	z_0

$$R'_r = R_r/p$$
$$R'_i = R_i - \bar{a}_{ik}R'_r \ (i \neq r), \qquad R'_q = R_q - q_k R'_r$$

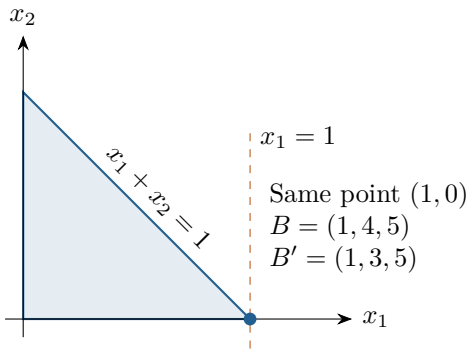
Basic	$\cdots$	x_j	$\cdots$	x_k	$\cdots$	RHS
$\vdots$		$\vdots$		$\vdots$		$\vdots$
x_{j_i}	$\cdots$	$\bar{a}_{ij} - \dfrac{\bar{a}_{ik}\bar{a}_{rj}}{p}$	$\cdots$	0	$\cdots$	$\bar{b}_i - \dfrac{\bar{a}_{ik}\bar{b}_r}{p}$
$\vdots$		$\vdots$		$\vdots$		$\vdots$
x_k	$\cdots$	$\dfrac{\bar{a}_{rj}}{p}$	$\cdots$	1	$\cdots$	$\dfrac{\bar{b}_r}{p}$
$\vdots$		$\vdots$		$\vdots$		$\vdots$
q'	$\cdots$	$q_j - \dfrac{q_k\bar{a}_{rj}}{p}$	$\cdots$	0	$\cdots$	$z_0 - \dfrac{q_k\bar{b}_r}{p}$

Figure 4: A general tableau pivot, with $i \neq r$ and $j \neq k$. Dots represent the remaining rows and columns. Blue marks the pivot row and the resulting unit column; orange marks the objective row R_q. The symbols $\bar{a}_{ij}, \bar{b}_i, q_j, z_0$ denote old values; R'_r is the normalized pivot row. The structural zeros and one are independent of the problem data.

Without degeneracy each pivot strictly increases the objective, so no basis can repeat; the finite number of bases implies termination. With degeneracy, $\theta^* = 0$ can change the basis without moving the point or increasing the objective. Some pivot rules may cycle.

Bland's rule prevents cycling in exact arithmetic: choose the eligible entering variable with the smallest *original variable index*; among minimum-ratio ties choose the leaving basic variable

with the smallest original index. Choosing only the topmost row in a tie is not this rule. Dantzig's rule with arbitrary tie breaking has no general finite-termination guarantee.



$$x_3 = 1 - x_1 - x_2, \quad x_4 = 1 - x_1, \quad x_5 = 1 - x_2$$

Figure 5: Degeneracy for $x_1 + x_2 \leq 1$, $x_1 \leq 1$, $x_2 \leq 1$, $x \geq 0$. At $(1, 0)$ three boundary conditions meet. Both indicated bases give $x = (1, 0, 0, 0, 1)$; exchanging the zero slack changes the basis only.

Floating-point implementations also require scale-aware feasibility and indicator tolerances. A theoretical zero may appear as a tiny signed number; exact-arithmetic pivot logic cannot simply be applied without numerical care.

6 Worked example: production planning

Task statement. A producer makes two products using three limited resources. The table gives the resource consumption per item, available amounts, and selling prices:

Quantity	Product 1	Product 2	Available
Resource 1 (kg/item; availability in kg)	2	1	10
Resource 2 (kg/item; availability in kg)	3	3	24
Resource 3 (kg/item; availability in kg)	3/2	0	6
Selling price (PLN/item)	3	2	—

Let x_1, x_2 be the quantities produced. Allow fractional quantities, assume constant resource consumption per item, and assume all produced items can be sold. Find the production plan that maximizes total revenue, using the basic simplex method. The model is

$$\max(3x_1 + 2x_2), \qquad 2x_1 + x_2 \le 10, \quad 3x_1 + 3x_2 \le 24, \quad \tfrac{3}{2}x_1 \le 6, \quad x_1, x_2 \ge 0.$$

Add slacks x_3, x_4, x_5 in that order, representing unused resources in kg, and use

$$A = \begin{bmatrix} 2 & 1 & 1 & 0 & 0 \\ 3 & 3 & 0 & 1 & 0 \\ 3/2 & 0 & 0 & 0 & 1 \end{bmatrix}, \quad b = \begin{bmatrix} 10 \\ 24 \\ 6 \end{bmatrix}, \quad c = \begin{bmatrix} 3 \\ 2 \\ 0 \\ 0 \\ 0 \end{bmatrix}.$$

Throughout the following tableaus the bottom row is $q = z - c$, with the current objective value in the rightmost cell. Constraint rows are labelled by their basic variables; all unlisted variables are zero at the current BFS.

6.1 Start: the slack basis

The ordered basis is $B = (3, 4, 5)$.

Basic	x_1	x_2	x_3	x_4	x_5	RHS
x_3	2	1	1	0	0	10
x_4	3	3	0	1	0	24
x_5	$\boxed{3/2}$	0	0	0	1	6
q	-3	-2	0	0	0	$z_0 = 0$

Dantzig's rule selects x_1, because $-3 < -2$. Ratios are $10/2 = 5$, $24/3 = 8$, and $6/(3/2) = 4$. Hence x_5 leaves, $\theta^* = 4$, and revenue increases by $3 \cdot 4 = 12$.

6.2 First pivot

The ordered basis is $B = (3, 4, 1)$. Divide the third row by $3/2$, then eliminate column x_1 from the first two rows and the indicator row. For example, $R_1 \leftarrow R_1 - 2R_3$ gives $x_2 + x_3 - \frac{4}{3}x_5 = 2$.

Basic	x_1	x_2	x_3	x_4	x_5	RHS
x_3	0	$\boxed{1}$	1	0	$-4/3$	2
x_4	0	3	0	1	-2	12
x_1	1	0	0	0	$2/3$	4
q	0	-2	0	0	2	$z_0 = 12$

The production point is $(4, 0)$. Enter x_2: its positive-column ratios are $2/1 = 2$ and $12/3 = 4$, while the third row has coefficient zero and is excluded. Thus x_3 leaves and revenue gains $2 \cdot 2 = 4$.

6.3 Second pivot

The ordered basis is $B = (2, 4, 1)$.

Basic	x_1	x_2	x_3	x_4	x_5	RHS
x_2	0	1	1	0	$-4/3$	2
x_4	0	0	-3	1	$\boxed{2}$	6
x_1	1	0	0	0	$2/3$	4
q	0	0	2	0	$-2/3$	$z_0 = 16$

At $(4, 2)$, enter x_5. A slack can enter the basis just like any other nonbasic variable. Its column is $(-4/3, 2, 2/3)^T$, so only rows 2 and 3 limit the step. Their ratios are $6/2 = 3$ and $4/(2/3) = 6$. The first row's negative coefficient means x_2 increases, not that its ratio should be tested. Thus x_4 leaves with step 3 and revenue gain $(2/3) \cdot 3 = 2$.

6.4 Third pivot and certificate

The ordered basis is $B = (2, 5, 1)$.

Basic	x_1	x_2	x_3	x_4	x_5	RHS
x_2	0	1	-1	$2/3$	0	6
x_5	0	0	$-3/2$	$1/2$	1	3
x_1	1	0	1	$-1/3$	0	2
q	0	0	1	$1/3$	0	$z_0 = 18$

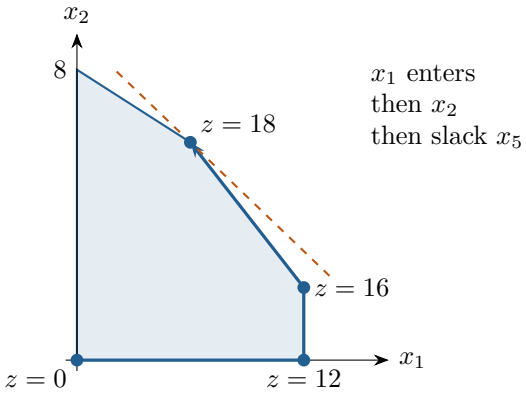
All indicators are nonnegative. Therefore

$$x^* = (2, 6, 0, 0, 3)^T, \qquad z^* = 18 \text{ PLN}.$$

The first two resources are exhausted, while 3 kg of resource 3 remain. The objective identity is an explicit certificate:

$$3x_1 + 2x_2 = 18 - x_3 - \tfrac{1}{3}x_4 \le 18. \tag{7}$$

Both nonbasic indicators are strictly positive, so this optimizer is unique. The same upper bound follows by adding the first resource constraint to one third of the second.



In	Out	Step	Revenue
x_1	x_5	4	12
x_2	x_3	2	16
x_5	x_4	3	18

Revenue in PLN.

Figure 6: The tableau sequence follows three edges of the production polygon. The final dashed line is $3x_1 + 2x_2 = 18$. Each nondegenerate pivot moves to a distinct adjacent vertex. The table summarizes the entering and leaving variables, steps, and resulting revenue.

7 Worked example: an unbounded objective

Task statement. Apply the primal simplex method to the following problem. If the objective is unbounded above, give an improving feasible ray:

$$\max(3x_1 + x_2 - 3x_3 - x_4), \quad \begin{cases} x_1 + 2x_2 - x_3 - x_4 \le 4, \\ 2x_1 - 2x_2 + 3x_3 - 3x_4 \le 9, \\ x_1, x_2, x_3, x_4 \ge 0. \end{cases}$$

Add slacks x_5, x_6. The initial basis is $(5, 6)$ and the tableau is

Basic	x_1	x_2	x_3	x_4	x_5	x_6	RHS
x_5	$\boxed{1}$	2	-1	-1	1	0	4
x_6	2	-2	3	-3	0	1	9
q	-3	-1	3	1	0	0	$z_0 = 0$

Enter x_1. The ratios 4 and $9/2$ select the first row, so x_5 leaves. After subtracting twice the first row from the second:

Basic	x_1	x_2	x_3	x_4	x_5	x_6	RHS
x_1	1	2	-1	-1	1	0	4
x_6	0	-6	5	-1	-2	1	1
q	0	5	0	-2	3	0	$z_0 = 12$

Now $q_4 = -2$ and the x_4 column is $(-1, -1)^T$. There is no positive entry and hence no leaving row. Set $x_4 = \theta$, keeping $x_2 = x_3 = x_5 = 0$. The equations give

$$x(\theta) = (4 + \theta, 0, 0, \theta, 0, 1 + \theta)^T, \quad d = (1, 0, 0, 1, 0, 1)^T, \quad \theta \ge 0. \tag{8}$$

Check directly: $Ad = 0$, $d \ge 0$, and $c^T d = 2$. The original constraint left-hand sides are 4 and $8 - \theta \le 9$, while the objective is $12 + 2\theta \to +\infty$. Thus unboundedness comes with a verifiable feasible ray, rather than just a failed ratio test.

This direction also spans an *extreme ray* of the standard-form recession cone $\{v : Av = 0, v \ge 0\}$. To see this, suppose $d = r + s$ with r, s in that cone. Since $d_2 = d_3 = d_5 = 0$, nonnegativity forces the same coordinates of r, s to be zero. The homogeneous equations then give $r_1 = r_4$ and $r_6 = r_4$, so $r = r_4 d$; the same holds for s. The decomposition can only use parallel nonnegative multiples of d.

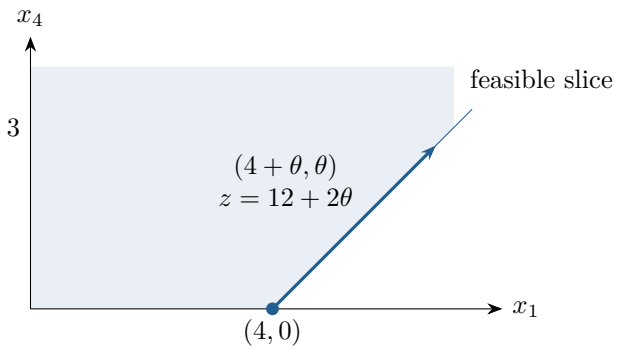


Figure 7: Slice $x_2 = x_3 = 0$ in the original four-variable problem. The shaded feasible region continues upward and to the right; the arrow lies on $x_1 - x_4 = 4$ and gives increasing objective.

8 Finding an initial feasible basis

A simplex step preserves feasibility; it cannot repair an infeasible starting basis. Before using the primal algorithm, convert the constraints to equalities and establish a BFS.

Constraint or situation	Equality form	Initial basis implication
$a^T x \le b$, $b \ge 0$	$a^T x + s = b$, $s \ge 0$	Slack column $+1$ can start basic with $s = b$.
$a^T x \ge b$, $b \ge 0$	$a^T x - s = b$, $s \ge 0$	Surplus column -1 alone gives $s = -b$; generally add an artificial column.
$a^T x = b$	Equality already present	Reuse a feasible basis if available; otherwise add an artificial column.
Equality with $b < 0$	Multiply the entire row by -1	Normalize RHS before introducing a positive artificial column.
Inequality with $b < 0$	Multiply by -1 and reverse inequality	Then choose slack or surplus for the new direction.

Slacks and surplus variables describe the original constraint. Artificial variables are temporary additions used to find feasibility; they must be zero in a solution of the original problem.

For $Ax \le b$, $b \ge 0$, slacks give the immediate basis $x = 0$, $s = b$. More generally, identity columns already present in an equality system can form an initial basis when their RHS values are nonnegative. An identity block with a negative basic value is not a feasible basis. Zero RHS values are allowed but produce degeneracy.

8.1 Phase I: a feasibility problem

After normalizing equality rows to $b \ge 0$, a universally available augmented system is

$$Ax + a = b, \qquad x \ge 0, \quad a \ge 0,$$

where each row has an artificial variable. It has the BFS $x = 0$, $a = b$. If some usable basic columns already exist, add artificials only to the rows that need them. Phase I solves

$$\min w = \sum_i a_i \quad \text{or, equivalently,} \quad \max(-w). \tag{9}$$

Always build its indicator row from its actual basic objective coefficients; the artificial variables may already be basic and cannot be treated as if their objective contribution were absent.

Proposition 8.1 (Phase I test)**.** *The original equality/nonnegativity system is feasible if and only if the Phase I optimum is $w^* = 0$.*

Proof. A feasible original x gives an augmented feasible point $(x, 0)$ with $w = 0$. Since $w \ge 0$, this is the minimum. Conversely, $w^* = 0$ and $a_i \ge 0$ force every artificial variable to vanish, so $Ax = b$, $x \ge 0$. □

Phase I is feasible by construction and its objective is bounded below by zero, so it has a finite attained minimum. A positive *intermediate* w says nothing about infeasibility; only a positive *optimal* w^* does.

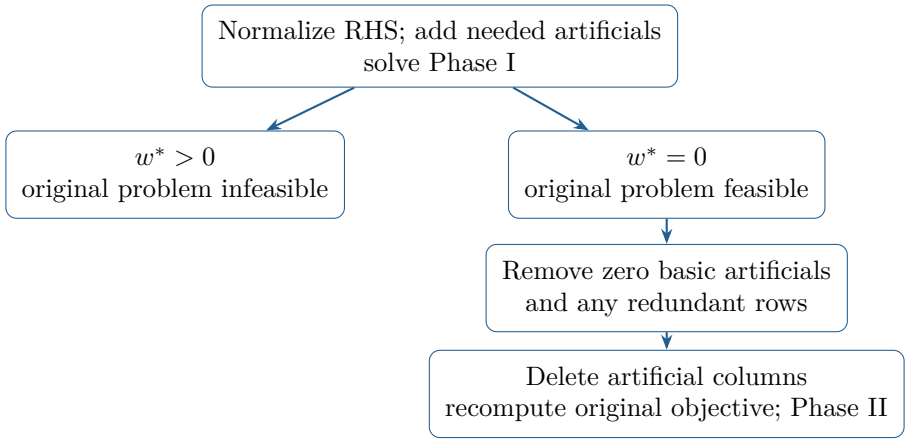


Figure 8: Two-phase initialization. Phase I establishes feasibility; Phase II optimizes the original objective.

8.2 Removing artificial variables from a zero optimum

At $w^* = 0$, a zero artificial variable may remain *basic*. Removing its column immediately would leave an incomplete basis. Inspect its transformed row restricted to original nonbasic columns:

- If a coefficient is nonzero, pivot that original variable into the row. The RHS is zero, so the point stays unchanged. For this cleanup exchange the coefficient may be positive or negative: this is a basis repair at zero, not an objective-improving positive-step ratio test.
- If every original coefficient in that row is zero, its original equality is $0 = 0$ after transformation. Delete that redundant row and its artificial basic variable.

Under the full-row-rank assumption on the original A, such a redundant row cannot occur; it matters when dependent rows have been retained. For instance, $x_1 = 1$ and $2x_1 = 2$ reduce to one independent equality. In contrast, $x_1 = 1$ and $2x_1 = 3$ are inconsistent, not a redundant pair. After cleanup, delete artificial columns and recompute q and z_0 using the *original* objective and the retained basis. Phase I indicators cannot certify optimality for Phase II.

8.3 A small infeasible example

Task statement. Use Phase I to decide whether $x \geq 2$, $x \leq 1$, $x \geq 0$ is feasible.

Introduce surplus s, slack t, and artificials:

$$x - s + a_1 = 2, \qquad x + t + a_2 = 1, \qquad x, s, t, a_1, a_2 \geq 0.$$

Subtracting the second equation from the first gives $a_1 = 1+s+t+a_2$. Therefore $w = a_1+a_2 \geq 1$. The augmented point $(x, s, t, a_1, a_2) = (1, 0, 0, 1, 0)$ attains $w = 1$, so $w^* = 1 > 0$ and the original problem is infeasible. This proves the Phase I result without relying on a numerical solver status.

9 Worked example: two different Phase I starts

Task statement. Solve the following problem by two-phase simplex, comparing a start with two artificial variables against one that reuses the original column a_2:

$$\max(-x_1 - 2x_2 - 3x_3 - 4x_4), \qquad \begin{cases} x_1 + x_2 + x_3 + x_4 = 1, \\ x_1 + x_3 - 3x_4 = \frac{1}{2}, \\ x_1, x_2, x_3, x_4 \geq 0. \end{cases} \tag{10}$$

The coefficient matrix has rank 2, but its original columns do not provide an immediate feasible identity basis. Compare an all-artificial start with a start that reuses the existing column $a_2 = (1, 0)^T$.

9.1 Start A: two artificial variables

Introduce artificial variables x_5, x_6 and maximize $-x_5 - x_6$ in Phase I:

$$x_1 + x_2 + x_3 + x_4 + x_5 = 1, \qquad x_1 + x_3 - 3x_4 + x_6 = \tfrac{1}{2}.$$

The basis $(5, 6)$ gives $x_5 = 1$, $x_6 = 1/2$, and $w = 3/2$. Use the Phase I coefficients $c^{\mathrm{I}} = (0, 0, 0, 0, -1, -1)^T$:

Basic	x_1	x_2	x_3	x_4	x_5	x_6	RHS
x_5	1	1	1	1	1	0	1
x_6	$\boxed{1}$	0	1	-3	0	1	$1/2$
q^{I}	-2	-1	-2	2	0	0	$z_{\mathrm{I}} = -3/2$

The most negative indicators tie for x_1 and x_3; select x_1 by smaller index. Ratios 1 and $1/2$ select row 2, so x_6 leaves.

Basic	x_1	x_2	x_3	x_4	x_5	x_6	RHS
x_5	0	1	0	$\boxed{4}$	1	-1	$1/2$
x_1	1	0	1	-3	0	1	$1/2$
q^{I}	0	-1	0	-4	0	2	$z_{\mathrm{I}} = -1/2$

Now x_4 enters and only row 1 has a positive coefficient. The step is $(1/2)/4 = 1/8$, and x_5 leaves. The Phase I objective increases by $4 \cdot (1/8) = 1/2$ to zero.

Basic	x_1	x_2	x_3	x_4	x_5	x_6	RHS
x_4	0	$1/4$	0	1	$1/4$	$-1/4$	$1/8$
x_1	1	$3/4$	1	0	$3/4$	$1/4$	$7/8$
q^{I}	0	0	0	0	1	1	$z_{\mathrm{I}} = 0$

Both artificials are nonbasic and zero. Delete columns 5 and 6; retain $B = (4, 1)$. Recompute the objective using $c^{\mathrm{II}} = (-1, -2, -3, -4)^T$:

Basic	x_1	x_2	x_3	x_4	RHS
x_4	0	$1/4$	0	1	$1/8$
x_1	1	$3/4$	1	0	$7/8$
q^{II}	0	$1/4$	2	0	$z_{\mathrm{II}} = -11/8$

All Phase II indicators are nonnegative, so this start has already reached the original optimum. Phase I happened to find a basis optimal for Phase II; this is not guaranteed in general.

9.2 Start B: reuse an original basic column

Original problem (recap). Solve the same program, now using only one artificial variable:

$$\max(-x_1 - 2x_2 - 3x_3 - 4x_4), \qquad \begin{cases} x_1 + x_2 + x_3 + x_4 = 1, \\ x_1 + x_3 - 3x_4 = \frac{1}{2}, \\ x_1, x_2, x_3, x_4 \geq 0. \end{cases}$$

Keep x_2 basic in row 1 and introduce only $x_5 = a$ in row 2:

$$x_1 + x_2 + x_3 + x_4 = 1, \qquad x_1 + x_3 - 3x_4 + x_5 = \tfrac{1}{2}.$$

Start with $B = (2, 5)$ and maximize $-x_5$.

Basic	x_1	x_2	x_3	x_4	x_5	RHS
x_2	1	1	1	1	0	1
x_5	$\boxed{1}$	0	1	-3	1	1/2
q^{I}	-1	0	-1	3	0	$z_{\mathrm{I}} = -1/2$

Enter x_1 and remove x_5 after step 1/2:

Basic	x_1	x_2	x_3	x_4	x_5	RHS
x_2	0	1	0	4	-1	1/2
x_1	1	0	1	-3	1	1/2
q^{I}	0	0	0	0	1	$z_{\mathrm{I}} = 0$

Delete column 5. The feasible original point is $(1/2, 1/2, 0, 0)^T$, but recomputing its *original* objective gives

Basic	x_1	x_2	x_3	x_4	RHS
x_2	0	1	0	$\boxed{4}$	1/2
x_1	1	0	1	-3	1/2
q^{II}	0	0	2	-1	$z_{\mathrm{II}} = -3/2$

The negative indicator for x_4 requires a Phase II pivot. Enter x_4, remove x_2, and take step 1/8. Dividing row 1 by 4 and adding three copies to row 2 produces exactly the final Phase II tableau from Start A. Therefore both starts yield

$$x^* = (7/8, 0, 0, 1/8)^T, \qquad z^* = -11/8. \tag{11}$$

Its indicator identity gives a full hand certificate:

$$-x_1 - 2x_2 - 3x_3 - 4x_4 = -\tfrac{11}{8} - \tfrac{1}{4}x_2 - 2x_3 \leq -\tfrac{11}{8}. \tag{12}$$

The two nonbasic indicators are positive, so the optimum is unique.

9.3 Geometry of the feasible equality system

Eliminate x_1, x_2 from Equation (10):

$$x_1 = \tfrac{1}{2} - x_3 + 3x_4, \qquad x_2 = \tfrac{1}{2} - 4x_4.$$

Nonnegativity gives $0 \le x_4 \le 1/8$ and $0 \le x_3 \le 1/2 + 3x_4$. These two coordinates parameterize the whole feasible set, and

$$z = -\tfrac{3}{2} + x_4 - 2x_3.$$

The maximum is at the largest allowed x_4 and smallest allowed x_3. This independent calculation confirms the tableau result.

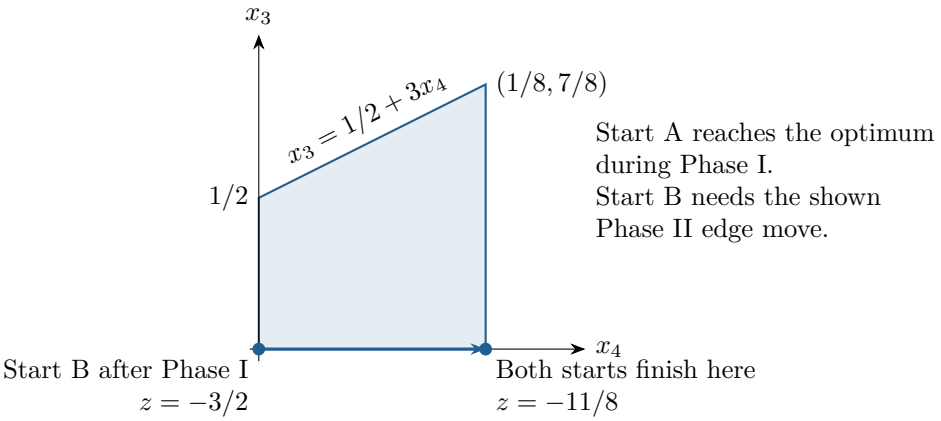


Figure 9: The four-variable equality system is a quadrilateral in coordinates (x_4, x_3). Its lower edge has increasing objective toward the unique optimum.

Initialization	Artificial variables	Phase I pivots	Phase II pivots
Start A	2	2	0
Start B	1	1	1

Reusing an existing basis column saves an artificial variable here. It does not imply that every such reuse minimizes the total number of simplex pivots.

10 The Big M method and its limitations

After normalizing the RHS and introducing needed artificials, the maximization penalty formulation is

$$\max(c^T x - M \textstyle\sum_i a_i), \qquad Ax + a = b, \quad x, a \ge 0. \tag{13}$$

Here the displayed form adds one artificial per row; a mixed starting basis can use fewer. Positive artificial values are penalized, but M must dominate the competing original objective for the intended feasibility priority. A rigorous symbolic interpretation compares objective expressions lexicographically: first minimize $\sum a_i$, then optimize $c^T x$ among points with the minimum artificial sum. This recovers the two-phase priority. When comparing $\alpha M + \beta$ symbolically, compare α first and then β; an arbitrary large floating-point constant is not the same operation.

10.1 A symbolic Big M calculation for the equality example

Task statement. Solve the following original problem using a dominant symbolic Big M penalty:

$$\max(-x_1 - 2x_2 - 3x_3 - 4x_4), \qquad \begin{cases} x_1 + x_2 + x_3 + x_4 = 1, \\ x_1 + x_3 - 3x_4 = \frac{1}{2}, \\ x_1, x_2, x_3, x_4 \geq 0. \end{cases}$$

Introduce $x_5, x_6 \geq 0$ with

$$x_1 + x_2 + x_3 + x_4 + x_5 = 1, \qquad x_1 + x_3 - 3x_4 + x_6 = \tfrac{1}{2},$$

and maximize $-x_1 - 2x_2 - 3x_3 - 4x_4 - M(x_5 + x_6)$. The combined coefficient vector is $c^M = (-1, -2, -3, -4, -M, -M)^T$. Its initial basis $(5, 6)$ gives

$$q^M = (1 - 2M,\ 2 - M,\ 3 - 2M,\ 4 + 2M,\ 0,\ 0)^T, \qquad z_M = -3M/2.$$

For symbolic dominant M, x_1 enters and x_6 leaves after step $1/2$. The constraint rows are exactly the first transformed rows of Start A; only their objective coefficients differ. Recompute with the combined objective at $B = (5, 1)$:

$$q^M = (0,\ 2 - M,\ 2,\ 7 - 4M,\ 0,\ 2M - 1)^T, \qquad z_M = -(M + 1)/2.$$

Now x_4 enters and x_5 leaves after step $1/8$. At $B = (4, 1)$,

$$q^M = (0,\ 1/4,\ 2,\ 0,\ M - 7/4,\ M + 3/4)^T, \qquad z_M = -11/8.$$

Both artificials are zero and every indicator is nonnegative for dominant M. This certifies the same original optimum $(7/8, 0, 0, 1/8)^T$ as the two-phase method. For this *specific* problem, the same Dantzig pivot sequence and strict nonbasic indicator certificate also work with any finite $M > 7/4$. For example $M = 10$ gives penalty-objective values -15, $-11/2$, and $-11/8$. The threshold depends on the data; it is not a rule for other LPs.

10.2 A hand counterexample to an insufficient penalty

Task statement. For $K > M > 0$, compare the original problem $\max(-Kx)$ subject to $x = 1$, $x \geq 0$ with its Big M formulation. Does its penalized optimum necessarily have artificial variable $a = 0$?

The original problem is feasible, with unique value $-K$. Its augmented problem is

$$\max(-Kx - Ma), \qquad x + a = 1, \quad x, a \geq 0, \qquad K > M > 0.$$

Substitute $a = 1 - x$:

$$-Kx - M(1 - x) = -M + (M - K)x, \qquad 0 \leq x \leq 1.$$

The maximum is at $x = 0, a = 1$, with value $-M > -K$. A positive artificial variable at this penalized optimum does *not* prove the original system infeasible. For $K = 100$, $M = 10$, the penalized optimum is -10 while the original feasible point has value -100.

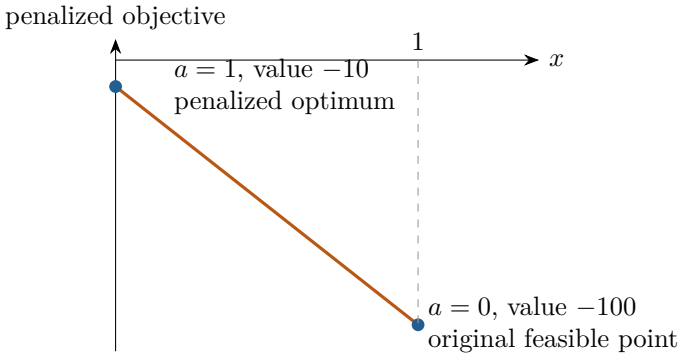


Figure 10: With $K = 100$ and $M = 10$, the penalty rewards keeping an artificial variable because avoiding the original cost is more valuable.

10.3 What a penalized solve can certify

A *finite optimal* augmented solution with $a = 0$ certifies an original optimum even for finite M: every original feasible point also belongs to the augmented system and has the same objective there. In contrast, interpreting positive artificials requires a justified feasibility priority.

An unbounded penalized problem with a positive artificial variable on a ray also does not alone prove original infeasibility. For example, $x_1 - x_2 + a = 1$ and objective $x_1 - Ma$ admit the augmented ray $(x_1, x_2, a) = (\theta, \theta, 1)$ with increasing objective. The original $a = 0$ system is nevertheless feasible and unbounded, using $(x_1, x_2) = (1+\theta, \theta)$. To certify original unboundedness, establish an original feasible point and an improving ray whose artificial components are zero.

Aspect	Two-phase method	Big M method
Feasibility priority	Separate minimization of artificial sum	Dominant symbolic penalty or justified numerical M
Objective rows	Recompute the original row after Phase I	Original and penalty coefficients share a row
Infeasibility test	$w^* > 0$ at Phase I optimum	Positive artificials require a valid dominance argument
Numerical issue	Ordinary feasibility/optimality tolerances	Large coefficients can also amplify scaling and cancellation problems

For hand calculations, both formulations explain artificial variables. For a transparent numerical demonstration, two separate phases avoid inventing a penalty size. A simplex tableau is a teaching representation; efficient solvers usually maintain basis factorizations instead of repeatedly forming full inverses.

11 From a full tableau to revised simplex

The algebraic inverse A_B^{-1} explains a basis transformation. An implementation can obtain the same information by solving linear systems:

$$A_B x_B = b, \qquad A_B^T y = c_B, \qquad q_j = a_j^T y - c_j.$$

After selecting an entering k with $q_k < 0$, solve $A_B u = a_k$. Apply the same ratio test to $(x_B)_i / u_i$ over rows with $u_i > 0$. If none exist, the direction certifies unboundedness. This is *revised simplex*: it computes the needed quantities without storing the entire transformed matrix $A_B^{-1} A$.

With identical exact pivot choices, it follows the same bases as the tableau method and has the same degeneracy issues.

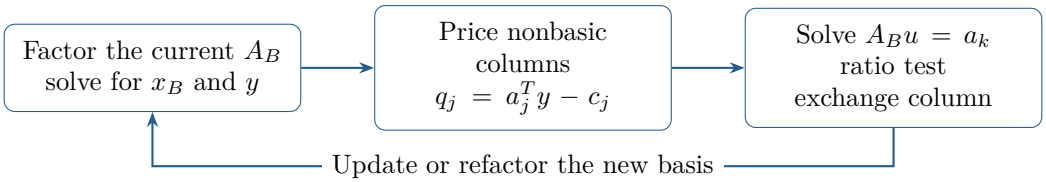


Figure 11: The revised formulation retains the mathematical basis exchange. Pricing selects an improving column or establishes that all indicators are nonnegative.

Sparse factorizations and their updates make this useful for large models; periodic refactorization limits accumulated numerical error. Presolve can remove fixed variables and redundant rows, but solutions and certificates must be mapped back to the original model. A small equation residual alone does not guarantee accurate coordinates when A_B is ill-conditioned. Scale variables and rows sensibly, distinguish numerical tolerances from exact sign tests, and verify original feasibility as well as dual indicators.

12 Checking the hand calculations in MATLAB

The companion script is `matlab/exercises/lecture_03_checks.m`. Run it from the repository root:

```
run('matlab/exercises/lecture_03_checks.m')
```

It uses base MATLAB linear solves and assertions, without Optimization Toolbox. It checks the stated bases and pivot choices rather than implementing a general purpose LP solver. The same matrices and objective coefficients are used in the hand calculations and the script.

A basis calculation is just two linear solves:

```
AB = A(:, B);
Abar = AB \ A;
bbar = AB \ b;
q = Abar' * c(B) - c;
x = zeros(size(A, 2), 1);
x(B) = bbar;
value = c' * x;
```

Here B is an ordered row vector of indices; b, c, x, q are column vectors. The ratio test deliberately discards zero and negative column coefficients:

```
u = Abar(:, entering);
rows = find(u > tol);
if isempty(rows)
    direction = zeros(size(A, 2), 1);
    direction(B) = -u;
    direction(entering) = 1;
else
    ratios = bbar(rows) ./ u(rows);
    step = min(ratios);
end
```

This fragment illustrates the algebra; the companion script additionally checks entering indicators, leaving choices, feasibility, tableau row operations, and agreement with the next independently recomputed basis. It uses an absolute tolerance 10^{-10} appropriate for these small, modestly

scaled inputs. A generic solver needs scale-aware tolerances and a strategy for ill-conditioned bases. Exact fractions in the notes are not rounded numerical results.

Check	Expected result	Interpretation
Production basis sequence	Revenue $0, 12, 16, 18$ PLN	Three improving pivots
Production final point	$(2, 6, 0, 0, 3)^T$	Slacks in kg
Unbounded example	$Ad = 0$, $d \geq 0$, $c^T d = 2$	Objective $12 + 2\theta$
Two-phase Start A	$w : 3/2, 1/2, 0$	Two feasibility pivots
Two-phase Start B	$w : 1/2, 0$	One feasibility pivot
Original equality problem	$(7/8, 0, 0, 1/8)^T$, $z = -11/8$	Same optimum
Infeasible inequalities	$w^* = 1$	Infeasible
Big M equality solve	$M = 10$, $z = -11/8$, $a = 0$	Original optimum
Finite penalty example	$x = 0, a = 1$, value -10	$M = 10 < K = 100$

13 Study questions and common mistakes

Before doing another tableau, check these distinctions:

Mistake	Correct check
Every basic solution is feasible	Verify both $Ax = b$ and $x \geq 0$.
A zero variable always means degeneracy	Only a zero *basic* coordinate establishes degeneracy.
An unbounded set means an unbounded LP	Exhibit a feasible ray with *positive* objective slope.
Test every ratio in the entering column	Include only strictly positive coefficients.
A zero indicator proves another optimum	Verify a positive feasible step or otherwise establish a distinct optimal point.
A simplex pivot always moves the point	A zero step can change only the basis.
A positive Phase I value proves infeasibility	Complete Phase I to its optimum first.
Delete all artificial columns immediately	First replace zero basic artificials or remove redundant rows.
Reuse Phase I indicators in Phase II	Recompute with the original objective coefficients.
Any large M is enough	Justify dominance; positive artificial values alone do not certify infeasibility.

1. In the production example, compute the initial pivot if x_2 enters first. How does its vertex path differ from Dantzig's path?
2. Starting at the final production basis, explain why increasing either nonbasic slack cannot improve revenue.
3. In the unbounded example, verify the entire ray in the original inequalities, without using the transformed tableau.
4. In Start B of the two-phase example, derive $q_4^{\mathrm{II}} = -1$ directly from $c_B^T \bar{a}_4 - c_4$.
5. For the finite penalty counterexample, characterize all optimizers when $M < K$, $M = K$, and $M > K$.

Answers for checking. (1) Ratios 10 and 8 give step 8, so x_4 leaves and the point is $(0, 8)$; then x_1 enters with step 2 and x_3 leaves, reaching $(2, 6)$. (2) Equation (7) subtracts $x_3 + x_4/3$

from 18. (3) The two left-hand sides are 4 and $8 - \theta$. (4) $(-2)4 + (-1)(-3) - (-4) = -1$. (5) Respectively $(x, a) = (0, 1)$, the whole segment $x + a = 1$ with $x, a \geq 0$, and $(x, a) = (1, 0)$.

14 Sources and editorial remarks

Lecture highlighting. Cream boxes identify selected statements, formulas and problem data recorded in the supplied current lecture sources. The course map in `docs/course-map-2026-27.md` gives the page destinations. Added hand proofs, qualifications and solution steps are not claimed as recorded lecture content.
Primary source: E. Pawelec, `s4_w3.pdf`, PDF pages 1–12, printed slides 1–46.
Source date: Stored in the folder labelled 2020; PDF metadata records creation on 2019-10-17, which does not establish the delivery date.
Current course alignment: `ProgMat_w1.pdf`, pp. 32–39, confirms simplex and initialization coverage. Pages 36–39 contain the same equality example developed in Section 9. Sheet 1, group 1 requires the methods practiced in `01-primal-simplex.tex`.
Editorial scope: The complete mathematical content is reorganized in English. The worked tableaus retain the supplied data and both Phase I starts. Proofs, counterexamples, study questions, redrawn figures, and the MATLAB companion are additional explanations.

PDF pages	Printed slides	Material in these notes
1	1–4	Section 1; geometric interpretation
2–3	5–12	Sections 2–3; candidate solutions
4–6	13–23	Sections 3–5; simplex derivation
6–7	24–28	Section 6; full production tableaus
8	29–31	Section 7; unbounded example
8–9	32–36	Section 8; initialization and two phases
10	37–38	Section 10; penalty method
10–11	39–44	Section 9; two initialization variants
12	45–46	Section 4.4; pivot formulas

The Polish prompt in Section 6 is from slide 24. Candidate vectors in Section 2 are from slides 8 and 12. The unbounded ray is fully derived here from the example and question on slides 29–31. All figures are editable explanatory TikZ redrawings; the ratio test sketch, degeneracy example, and finite-penalty graph use added illustrative data. The equations and tableaus on slides 39–44 were inspected visually to confirm nonnegativity of all four original variables and both artificial-variable conventions.

Current-source qualifications. In `ProgMat_w1.pdf`, p. 34, an easily usable basic column should have a positive entry in its pivot row and zero entries in the other rows; the printed indices instead test other entries in a row. Normalize a positive column before reading its basic value. The penalty claims on page 35 require the Big M qualifications in Section 10. The image-based tableaus on pp. 37–39 were checked visually: the all-artificial path already developed here has Phase I pivots entering x_1 and then x_4, and Phase II needs no further pivot for that path. This does not imply that Phase II is generally unnecessary.

Qualifications of the mathematical claims. Section 3 distinguishes a sufficient indicator certificate from necessity for a nondegenerate BFS, qualifying the equivalence stated on slide 14. It also requires an actual positive feasible step to infer a distinct optimizer from a zero indicator. Section 8 explains zero-basic-artificial cleanup and redundant rows explicitly. Section 10 qualifies the classification on slide 38: an insufficient finite penalty and an augmented ray carrying positive artificial values do not by themselves establish original infeasibility.

Supplementary checks. Section 11 and Figure 11 connect the existing pivot equations to implementation. The terminology was checked against MIT OCW, *The Simplex Method II*, Lecture

6, PDF pp. 2–3. This is additional study material with unknown current-course alignment; no new solver implementation or MATLAB execution is claimed for this section.

MIT OCW 15.081J/6.251J, Lecture 5 (Fall 2009), PDF p. 2 and pp. 5–6, was consulted for the nondegeneracy qualification and anti-cycling rule. Lecture 7, PDF pp. 2–4, was consulted for removal of zero basic artificials and the transition to Phase II. These are supplementary references, not evidence of current course coverage; the source inventory records external storage status.

Verification. All displayed worked-example tableaus were independently recomputed using exact rational Gauss–Jordan elimination, with constraint residuals checked exactly. Their objective and indicator rows agree with the hand derivations. The installed R2026b static code analyzer checked the MATLAB companion with no diagnostics. On 2026-10-04 the user ran the companion in a normal terminal and supplied its output: all numerical assertions passed. The reported production sequence, unbounded-ray slope, two-phase optimum, and infeasibility result agree with Section 12. The run did not print a version; an earlier user run reported MATLAB R2026b. The supplied terminal output is recorded in `docs/matlab-verification.md`. Polish terminology is maintained in `docs/glossary.md`.